Casimir Effect Between Plates with Oxide Film *

SHAO Cheng-gang(邵成刚), LIU Zhong-zhu(刘中柱), LUO Jun(罗俊)
Department of Physics, Huazhong University of Science and Technology, Wuhan 430074

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The Casimir force between the plates with the oxide film in (3+1)-dimensional version is studied by means of the source theory. Theoretical analyses show that the Casimir force between the plates could be affected by the oxide film on the surface of the plate. Numerical simulation shows that the variation due to the oxide film will reach at the level of about 5% for the typical experimental parameters, which means that the oxide film effect should be considered in the high precision experiment.

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Casimir first pointed out that there is the attractive interaction between two neutral conducting plates, which is called the Casimir effect.[1] It is attributed to the change in the electromagnetic zero-point energy due to the presence of the plates. The experiments have been performed to verify the existence of the Casimir force in recent four decades,[2-5] and only Lamoreaux successfully observed this effect between a plate and a spherical surface with a precision torsion pendulum.[6] Although the measurement accuracy reported has reached in the order of 5%, it is still necessary to point out that the effect due to the oxide film on the plates should be counted in both the theory and the experiment. This is because the surface of the plate will oxidize as it is usually exposured in the air for a period of time after being manufactured. For simplification, let us consider the problem as shown in Fig 1. Two semi-infinite media with dielectric constants $\varepsilon_1(\omega)$ and $\varepsilon_4(\omega)$ are separated by two media with dielectric constants $\varepsilon_3(\omega)$ and $\varepsilon_2(\omega)$, where ε_1 and ε_4 represent the conducting plates when ε_1 and ε_4 intend infinite, ε_3 and ε_2 represent the oxide film on the plate and the vacuum, respectively. In this case, we need to calculate the Casimir effect between the dielectric plate and the oxide film in (3+1)-dimensional version. It is noted that the Casimir effect between dielectric plates was calculated by Lifshitz based on the fluctuation theory of the electromagnetic filed.[7] Later, it was derived by Schwinger *et al.*, with the source theory.[8] Considering the correlation of the vacuum fluctuation, the theories of Lifshitz and Schwinger were proved to be equivalent.[9] In this article, we calculate the Casimir effect with Schwinger's source theory.

To calculate the Casimir force acting on the surface at $z = 0$, we will consider the change in the energy of the system when the separation between media $\varepsilon_1(\omega)$ and $\varepsilon_3(\omega)$ is changed from $a - b$ to $a - b + \delta a$. Physically, the electromagnetic zero-point field fluctuations induce local dipoles in the dielectric, and these local dipoles induce the others dipoles in the broader range in the dielectric. So these randomly fluctuating dipoles are called polarization sources, which can be represented by $\boldsymbol{P}$. As a result of the spatial correlation of the fluctuations, the interaction between these dipoles leads to the Casimir force. The effective product of polarization sources $\boldsymbol{P}$ for zero temperature is written as follows

$$\mathrm{i}\boldsymbol{P}(x)\boldsymbol{P}(x') = \boldsymbol{I}\delta\varepsilon\delta(x - x'), \tag{1}$$

where x denotes a four-dimensional coordinate $(\boldsymbol{r}, t)$, $\boldsymbol{I}$ is a unit tensor or dyadic, and $\delta\varepsilon$ is an infinite small change in the dielectric constant.

The linear relation between source and filed is

$$\boldsymbol{E}(x) = \int (\mathrm{d}x')\boldsymbol{\Gamma}(x, x') \cdot \boldsymbol{P}(x'), \tag{2}$$

and the change in the energy is

$$\delta E = \frac{\mathrm{i}}{2}\int (\mathrm{d}\boldsymbol{r})\frac{\mathrm{d}\omega}{2\pi}\delta\varepsilon(\boldsymbol{r}, \omega)\boldsymbol{\Gamma}_{kk}(\boldsymbol{r}, \boldsymbol{r}, \omega), \tag{3}$$

where $\boldsymbol{\Gamma}$ is Green's dyadic and repeated indices imply summation.[8] For a variation in the position of the interface between media ε_1 and ε_2, which means that the polarization sources are only on the plane $z = 0$, the change in the dielectric constant is

$$\delta\varepsilon(z) = -\delta a(\varepsilon_1 - \varepsilon_2)\delta(z). \tag{4}$$

Therefore, the force per unit area on the plane $z = 0$ is

$$f = \frac{\mathrm{i}}{2}\int \frac{\mathrm{d}w}{2\pi}\frac{(\mathrm{d}\boldsymbol{k}_\perp)}{(2\pi)^2}(\varepsilon_1 - \varepsilon_2) \cdot \left[\omega^2 g^E + \frac{k^2}{\varepsilon_1\varepsilon_2}g^H + \frac{1}{\varepsilon}\frac{\partial}{\partial z}\frac{1}{\varepsilon'}\frac{\partial}{\partial z'}g^H\right]\Bigg|_{z=z'=0}, \tag{5}$$

where the "transverse electric" and "transverse magnetic" Green's functions satisfy the following equa-

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tions,

$$\left[-\frac{\partial^2}{\partial z^2}+k^2-\omega^2\varepsilon\right]g^E(z,z')=\delta(z-z'), \quad (6a)$$

$$\left[-\frac{\partial}{\partial z}\frac{1}{\varepsilon}\frac{\partial}{\partial z}+\frac{k^2}{\varepsilon}-\omega^2\right]g^H(z,z')=\delta(z-z'). \quad (6b)$$

Introducing the quantity $K_i^2=k^2-\omega^2\varepsilon_i$ $(i=1,2,3,4)$, we get the solution for g^E in the region of $z,\ z'>0$

$$g^E(z,z')=\frac{\mathrm{e}^{-K_1|z-z'|}+r\mathrm{e}^{-K_1(z+z')}}{2K_1}, \quad (7a)$$

where the reflection coefficient r is given by

$$\begin{aligned}1+r=&\frac{2K_1}{K_1+K_2}+\frac{4K_1K_2}{K_2^2-K_1^2}\\&\cdot\left(\frac{K_3+K_4}{K_3-K_4}\mathrm{e}^{2K_3b}+\frac{K_2+K_3}{K_2-K_3}\right)\\&\cdot\left[\left(\frac{K_1+K_3}{K_2-K_3}\frac{K_2+K_1}{K_2-K_1}\mathrm{e}^{2K_2(a-b)}-1\right)\right.\\&\cdot\frac{K_3+K_4}{K_3-K_4}\mathrm{e}^{2K_3b}+\frac{K_2+K_1}{K_2-K_1}\mathrm{e}^{2K_2(a-b)}\\&\left.-\frac{K_2+K_3}{K_2-K_3}\right]^{-1}. \quad (7b)\end{aligned}$$

The magnetic Green's function g^H has the same form but with the replacement $K_i\to K_i/\varepsilon_i\equiv K_i'(i=1,2,3,4)$. So the force per unit area on the plane $z=0$ is finally obtained as follows

$$\begin{aligned}f_{z=0}=&\frac{\mathrm{i}}{2}\int\frac{\mathrm{d}\omega}{2\pi}\frac{(\mathrm{d}k_x\mathrm{d}k_y)}{(2\pi)^2}2K_2\\&\cdot\left\{\left(\frac{K_3+K_4}{K_3-K_4}\mathrm{e}^{2K_3b}+\frac{K_2+K_3}{K_2-K_3}\right)\right.\\&\cdot\left[\left(\frac{K_2+K_3}{K_2-K_3}\frac{K_2+K_1}{K_2-K_1}\mathrm{e}^{2K_2(a-b)}-1\right)\right.\\&\cdot\frac{K_3+K_4}{K_3-K_4}\mathrm{e}^{2K_3b}+\frac{K_2+K_1}{K_2-K_1}\mathrm{e}^{2K_2(a-b)}\\&\left.-\frac{K_2+K_3}{K_2-K_3}\right]^{-1}\\&+\left(\frac{K_3'+K_4}{K_3'-K_4'}\mathrm{e}^{2K_3b}+\frac{K_2'+K_3'}{K_2'-K_3'}\right)\\&\cdot\left[\left(\frac{K_2'+K_3'}{K_2'-K_3'}\frac{K_2'+K_1'}{K_2'-K_1'}\mathrm{e}^{2K_2(a-b)}-1\right)\right.\\&\cdot\frac{K_3'+K_4'}{K_3'-K_4'}\mathrm{e}^{2K_3b}+\frac{K_2'+K_1'}{K_2'-K_1'}\mathrm{e}^{2K_2(a-b)}\\&\left.\left.-\frac{K_2'+K_3'}{K_2'-K_3'}\right]^{-1}\right\}, \quad (8)\end{aligned}$$

where we have remained the terms reference to the parameters a and b which give the Casimir force. It is noted that the first square bracket comes from the $\boldsymbol{E}$ polarization and the second from the $\boldsymbol{H}$ polarization.

Let $\omega=\mathrm{i}\xi$ and $K_2=p\xi\varepsilon_2$, so $K_i^2=k^2+\xi^2\varepsilon_i$ and Eq. (8) can be rewritten in the form as follows

$$\begin{aligned}f_{z=0}=&-\frac{1}{2\pi^2\sqrt{\varepsilon_2}}\int_0^\infty\mathrm{d}\xi\cdot\xi^3\int_1^\infty\mathrm{d}p\cdot p^2\\&\cdot\left\{\left(\frac{s_3+s_4}{s_3-s_4}\mathrm{e}^{2s_3\xi b}-\frac{s_3+p}{s_3-p}\right)\right.\\&\cdot\left[\left(\frac{s_3+p}{s_3-p}\frac{s_1+p}{s_1-p}\mathrm{e}^{2s_2\xi(a-b)}-1\right)\frac{s_3+s_4}{s_3-s_4}\mathrm{e}^{2s_3\xi b}\right.\\&\left.\cdot-\frac{s_1+p}{s_1-p}\mathrm{e}^{2s_2\xi(a-b)}+\frac{s_3+p}{s_3-p}\right]^{-1}\\&+\left(\frac{s_3+s_4\varepsilon_3/\varepsilon_4}{s_3-s_4\varepsilon_3/\varepsilon_4}\mathrm{e}^{2s_3\xi b}-\frac{s_3+p\varepsilon_3/\varepsilon_2}{s_3-p\varepsilon_3/\varepsilon_2}\right)\\&\cdot\left[\left(\frac{s_3+p\varepsilon_3/\varepsilon_2}{s_3-p\varepsilon_3/\varepsilon_2}\frac{s_1+p\varepsilon_1/\varepsilon_2}{s_1-p\varepsilon_1/\varepsilon_2}\mathrm{e}^{2s_2\xi(a-b)}-1\right)\right.\\&\cdot\frac{s_3+s_4\varepsilon_3/\varepsilon_4}{s_3-s_4\varepsilon_3/\varepsilon_4}\mathrm{e}^{2s_3\xi b}-\frac{s_1+p\varepsilon_1/\varepsilon_2}{s_1-p\varepsilon_1/\varepsilon_2}\mathrm{e}^{2s_2\xi(a-b)}\\&\left.\left.+\frac{s_3+p\varepsilon_3/\varepsilon_2}{s_3-p\varepsilon_3/\varepsilon_2}\right]^{-1}\right\}, \quad (9)\end{aligned}$$

where $s_i=\sqrt{\varepsilon_i/\varepsilon_2+p^2-1}$ $(i=1,3$ and $4)$. When $b\to0$ (or $b\to\infty$), then $\varepsilon_3\to\varepsilon_4$ (or $\varepsilon_3\to\varepsilon_2$), we can directly deduce the simple form of the Casimir force from Eq. (8) as same as in Refs. 7 and 8.

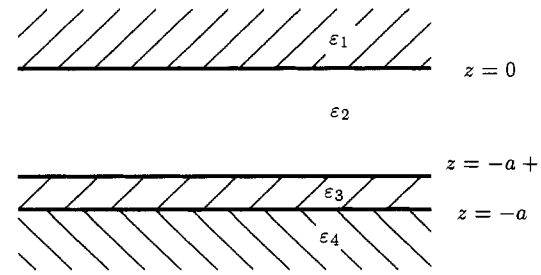


Fig. 1. Configuration of dielectric plates and film.

Above equation describes the Casimir force between the dielectric plates in (3+1)-dimensional version. If setting $\varepsilon_4=\varepsilon_1=\infty$, $\varepsilon_2=1$ and $\lambda=b/(a-b)$, which is the dimensionless film thickness, we can obtain the formula denoting the Casimir force between a perfectly conducting plate and a plate with the oxide film on the surface as follows

$$\begin{aligned}f_{z=0}=&-\frac{1}{32\pi^2(a-b)^4}\int_0^\infty\mathrm{d}x\int_1^\infty\mathrm{d}p\frac{x^3}{p^2}\\&\cdot\left\{\left[\frac{1+\frac{s_3+p}{s_3-p}\mathrm{e}^{-\lambda x s_3/p}}{\left(\frac{s_3+p}{s_3-p}\mathrm{e}^x-1\right)+\left(\mathrm{e}^x-\frac{s_3+p}{s_3-p}\right)\mathrm{e}^{-\lambda x s_3/p}}\right]\right.\\&\left.+\left[\frac{-1+\frac{s_3+\varepsilon_3 p}{s_3-\varepsilon_3 p}\mathrm{e}^{-\lambda x s_3/p}}{\left(\frac{s_3+\varepsilon_3 p}{s_3-\varepsilon_3 p}\mathrm{e}^x+1\right)-\left(\mathrm{e}^x+\frac{s_3+\varepsilon_3 p}{s_3-\varepsilon_3 p}\right)\mathrm{e}^{-\lambda x s_3/p}}\right]\right\}, \quad (10)\end{aligned}$$

where the Casimir force $f_{z=0}$ is only dependent on $(a-b)^4$ and λ. Figure 2 shows the dependence of the Casimir force, which is in the unit $-\hbar c\pi^2/[240(a-b)^4]$ representing the Casimir force between two perfectly conducting parallel plates, on the dimensionless film thickness of the dielectric $\varepsilon_3\lambda$.

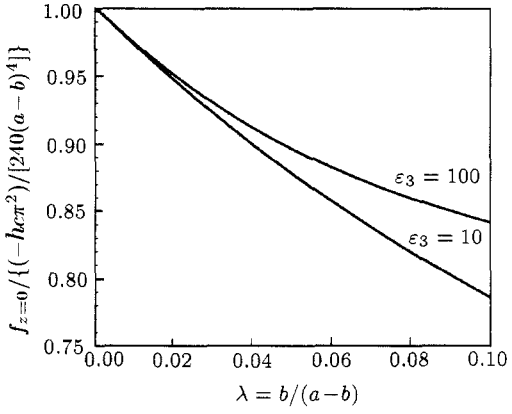


Fig. 2. Dependence of the function $f_{z=0}/\{-\hbar c\pi^2/[240(a-b)^4]\}$ on λ for $\varepsilon_4 = \varepsilon_1 = \infty$, $\varepsilon_2 = 1$. The two curves correspond to $\varepsilon_3 = 10$ and $\varepsilon_3 = 100$, in the region of the dielectric constant ε for usual metal oxide film.

It is noted that the thickness of the oxide film on the metal surface is in the order of nm at room temperature in the air. For example, the oxide film (Al_2O_3) on the aluminum plate is about 5 nm, and the dielectric constant ε of the oxide film Al_2O_3 is about 10 under the standard condition for a few days. In the experiment for measuring the Casimir effect, the distance a is usually about 0.5 μm, so the dimensionless parameter $\lambda = 1\%$, and the influence of the oxide film will reach at the level of about 2.5% as shown in Fig. 2. If the two aluminum plates are both oxidized with Al_2O_3, the influence will reach at 5%, which should be considered in a high precision experiment.

Instead of setting the polarization sources on the plane $z = 0$, the Casimir force between the plate and the film can be also given by the infinite small displacement of the other planes $z = -a+b$ and $z = -a$ equivalently, which is given by the change in the dielectric constant as follows

$$\delta\varepsilon(z) = -\delta a(\varepsilon_2-\varepsilon_3)\delta(z+a-b) - \delta a(\varepsilon_3-\varepsilon_4)\delta(z+a). \tag{11}$$

Therefore the force per unit area on the media ε_3 and ε_4 is

$$f_{z=-a+b} = \frac{\mathrm{i}}{2}\int \frac{\mathrm{d}\omega}{2\pi}\frac{(\mathrm{d}\boldsymbol{k}_\perp)}{(2\pi)^2}$$
$$\cdot\left\{(\varepsilon_2-\varepsilon_3)\left[\omega^2 g^E + \frac{k^2}{\varepsilon_2\varepsilon_3}g^H + \frac{1}{\varepsilon}\frac{\partial}{\partial z}\frac{1}{\varepsilon'}\frac{\partial}{\partial z'}g^H\right]\right|_{z=z'=-a+b}$$
$$+(\varepsilon_3-\varepsilon_4)\left[\omega^2 g^E + \frac{k^2}{\varepsilon_3\varepsilon_4}g^H + \frac{1}{\varepsilon}\frac{\partial}{\partial z}\frac{1}{\varepsilon'}\frac{\partial}{\partial z'}g^H\right]\Bigg|_{z=z'=-a}\Bigg\}. \tag{12}$$

Through complicated calculation, we obtain the following relations

$$(\varepsilon_2-\varepsilon_3)\omega^2 g^E\big|_{z=z'=-a+b} + (\varepsilon_3-\varepsilon_4)\omega^2 g^E\big|_{z=z'=-a} + (\varepsilon_1-\varepsilon_2)\omega^2 g^E\big|_{z=z'=0} = K_4 - K_1, \tag{13a}$$

$$(\varepsilon_2-\varepsilon_3)\left[\frac{k^2}{\varepsilon_2\varepsilon_3}g^H + \frac{1}{\varepsilon}\frac{\partial}{\partial z}\frac{1}{\varepsilon'}\frac{\partial}{\partial z'}g^H\right]\Bigg|_{z=z'=-a+b} + (\varepsilon_3-\varepsilon_4)\left[\frac{k^2}{\varepsilon_3\varepsilon_4}g^H + \frac{1}{\varepsilon}\frac{\partial}{\partial z}\frac{1}{\varepsilon'}\frac{\partial}{\partial z'}g^H\right]\Bigg|_{z=z'=-a} + (\varepsilon_1-\varepsilon_2)\left[\frac{k^2}{\varepsilon_1\varepsilon_2}g^H + \frac{1}{\varepsilon}\frac{\partial}{\partial z}\frac{1}{\varepsilon'}\frac{\partial}{\partial z'}g^H\right]\Bigg|_{z=z'=0} = K_4 - K_1. \tag{13b}$$

Since the term $K_4 - K_1$ is independent of a and b, we can discard it by regularization. So we finally get

$$f_{z=0} + f_{z=-a+b} = 0, \tag{14}$$

which means the Casimir force acting on the plane $z = 0$ and that on the plane $z = -a+b$ have the same magnitude but inverse direction. Moreover, that the different positions of sources give the same force means the symmetry of the theory for exchanging positions of sources.

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